Project Title: Driving ROI: Automotive Digital Retailing & Profitability Analysis

🏢 Project Overview

This project bridges the gap between front-line internet sales and backend data engineering. Using a synthetic dataset representing 500 digital leads, this analysis evaluates the performance of the "SmartPath" digital pipeline, moving beyond vanity volume metrics to quantify real profitability and sales efficiency.

🛠️ Technical Stack

- **Database**: SQLite (Relational Schema Design)
- **Programming**: Python (Pandas, Seaborn, Matplotlib)
- **Data Generation**: Faker (for realistic synthetic interaction logs)
- **Analysis**: SQL (Aggregate functions, Joins, Grouping)

📊 Key Insights

- **Profitability Trends**: Identified which lead sources (e.g., SmartPath vs. Third-Party) yield the highest Front-End Gross Profit.
- **Efficiency Metrics**: Analyzed the correlation between sales effort (touchpoints per deal) and conversion success.
- **Operational Intelligence**: Provided a blueprint for dealership management to identify high-potential traffic and optimize BDC/Internet Sales workflows.

🚀 Why This Matters

As an automotive professional, I know that success isn't just about the number of calls made—it's about the strategic application of effort. This project demonstrates my ability to take raw CRM interaction data and translate it into clear, actionable business intelligence that improves the bottom line.